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B.0 1D Calculus Review

0.1: differentiation

- The avg. rate of Δ from $x \to x+h \Rightarrow \frac{f(x+h)-f(x)}{h}$
- The derivative is the instantaneous rate of change $\Rightarrow \lim_{h \to 0} \frac{f(x+h)-f(x)}{h} = \frac{df}{dx}$

Eg. $\frac{d}{dx} x^2 = 2x$

- Some key formulas —

1. $\frac{d}{dx} f(x) = f'(x)$
2. $\frac{d}{dx} a = 0 \Rightarrow$ for constant a
3. $\frac{d}{dx} x^n = nx^{n-1}$
4. $\frac{d}{dx} (\ln x) = \frac{1}{x}$
5. $\frac{d}{dx} e^x = e^x + \frac{d}{dx}(x) = e^x$
6. $\frac{d}{dx} (f(x) + g(x)) = f'(x) + g'(x)$
7. $\frac{d}{dx} cf(x) = c\frac{d}{dx} f(x) = cf'(x)$

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Eg. 1. $\frac{d}{dx} 3x^2 = 6x$ 2. $\frac{d}{dx} x^3 = 3x^2$

3. $\frac{d}{dy} 3y^2 = 6y$ 4. $\frac{d}{dt}(5e^t + 3t^4) = 5e^t + 12t^3$

5. $\frac{d}{ds}\left(\frac{2}{s^3}\right) = \frac{d}{ds} 2s^{-3} = -6s^{-4} = \frac{-6}{s^4}$

- Chain Rule Theorem → If f and u are differentiable functions -

$$\boxed{\frac{d}{dx} f(u) = f'(u) \frac{du}{dx}}$$

Eg. $\frac{d}{dx} u^n = nu^{n-1} \frac{du}{dx}$ } impt. results to remember.

$\frac{d}{dx} e^u = e^u \frac{du}{dx}$

Eg. $\frac{d}{dt} e^{-2t+2t^2} = \left(e^{-2t+2t^2}\right) \times \frac{d}{dt}\left(-2t + 2t^2\right)$

$= \left(e^{-2t+2t^2}\right) \cdot \left(-2 + 4t\right)$

Eg. $\frac{d}{dt}(2x^2 + 5x + 3)^5 = 5(2x^2 + 5x + 3)^4 \cdot (4x+5)$

Eg. $\frac{d}{dt} e^{5e^t - 5 + 3t} = \left(e^{5e^t - 5 + 3t}\right) \cdot \left(5e^t + 3\right)$

- $$\boxed{\frac{d}{dx} 2^x = \frac{d}{dx} e^{x \ln(2)} = \ln(2) \cdot 2^x}$$

$\frac{d}{dx} a^x = \ln(a) \cdot a^x$

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- Product rule - $\boxed{\frac{d}{dx}(u \cdot v) = u\frac{dv}{dx} + \frac{du}{dx}v}$

for quotients - $\frac{d}{dx}\left(u \times \frac{1}{v}\right) = u\frac{d}{dx}\left(\frac{1}{v}\right) + \frac{1}{v}\frac{du}{dx}$

$= \frac{-uv' + u'v}{v^2}$

Eg. $\frac{d}{dx}\left(x^2 e^{3x}\right) = x^2 \frac{d}{dx} e^{3x} + e^{3x} \times 2x$

$= x^2 e^{3x} + e^{3x} \times 2x$

Eg. $\frac{d}{dx}\left(\frac{e^{-x}}{x^3}\right) = e^{-x}\frac{d}{dx}\left(\frac{1}{x^3}\right) + \frac{1}{x^3}\left(\frac{d}{dx}e^{-x}\right)$

$= e^{-x} \times -3x^{-4} + \frac{1}{x^3}\left(-1e^{-x}\right)$

$= \frac{-3e^{-x}}{x^4} + \left(-\frac{e^{-x}}{x^3}\right)$

- For absolute values

- $|x| = x$ if $x \geq 0$

$|x| = -x$ if $x < 0$

$$\boxed{\frac{d}{dx}|x| = \begin{cases} 1 & \text{if } x > 0 \\ -1 & \text{if } x < 0 \end{cases}}$$

for $x = 0$, $\frac{d}{dx}|x|$ is undefined.

- $\boxed{\frac{d}{dx}\cos(x) = -\sin(x)}$ $\boxed{\frac{d}{dx}\sin(x) = \cos(x)}$

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Eg. $\frac{d}{dx}\frac{(2x+5)}{x^2-3x+4} = \frac{(2x+5)(-1)(2x-3)}{x^2-3x+4} + \frac{2}{x^2-3x+4}$

- some extended examples -

· $\frac{d}{dx}\sin|x+2| = \frac{d}{dx}\sin(x+2) \qquad x+2>0$

$= \cos(x+2) \qquad x>-2$

$\frac{d}{dx}\sin|x+2| = \frac{d}{dx}\sin(-x-2) \qquad x+2<0$

$= -\cos(-x-2) \qquad x<-2$

Q.1a Differentiation Problems (1 - 10)

1. $\frac{d}{dx} e^{3x^2+5x} = e^{3x^2+5x}\cdot(6x+5)$

2. $\frac{d}{dx} 2^{2x} = \cancel{\ln(2x)\cdot 2^{2x}}$

$= \frac{d}{dx} e^{2x\ln 2} = 2\ln 2\, e^{2x\ln 2}$

$= 2\ln 2\,(2^{2x})$

3. $\frac{d}{dx}(x^2+5)e^{2x-3} = 2x\cdot e^{2x-3} + (x^2+5)(e^{2x-3})\times 2$

4. $\frac{d}{dx}\frac{x^2}{(1+x^2)} = \frac{2x}{(1+x^2)} + x^2\frac{d}{dx}\left(\frac{1}{(1+x^2)}\right)$

$\frac{d}{dx}\frac{1}{1+x^2} = \frac{d}{dx}(1+x^2)^{-1} = -1\,(1+x^2)^{-2}$

$= \frac{-1\times 2x}{(1+x^2)^2} = \frac{-2x}{(1+x^2)^2}$

$= \frac{2x}{(1+x^2)} + \frac{x^2(-2x)}{(1+x^2)^2} = \frac{2x}{(1+x^2)^2}$

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5. $\left.\frac{d}{dx}\frac{x}{1+x^2}\right|_{x=0} = 1 + x\frac{d}{dx}\frac{1}{1+x^2}$

$= 1$

6. $\left.\frac{d^2}{dx^2}\frac{x}{1+x^2}\right|_{x=0} = 1 \times \frac{d}{dx}\left(\frac{1}{1+x^2}\right) + x\frac{d}{dx}\left(\frac{1}{1+x^2}\right)$

$= 1$

7. $\frac{d}{dx}\left(1-3e^{-3x}\right) = \frac{d}{dx}(1) - \frac{d}{dx}\left(3e^{-3x}\right)$

$= +9e^{-3x}$

8. $\frac{d}{dx}\frac{x}{1+x^2} = \frac{1}{1+x^2} + \frac{x(-2x)}{(1+x^2)^3}$ | $2(1+x^2)^{-3}$ $\frac{-2\times 2x}{(1+x^2)^3}$

9. $\frac{d}{dx}\frac{3x-5}{(4+x)^3} = \frac{3}{(4+x)^3} + 3x-5\frac{d}{dx}(4+x)^{-3}$

$\frac{d}{dx}(4+x)^{-3} = -3(4+x)^{-4} = \frac{-3}{(4+x)^4}$

$= \frac{3}{(4+x)^3} + \frac{(3x-5)(-3)}{(4+x)^4}$

10. $\frac{d}{dt}e^{5e^t-5} = e^{5e^t-5}\times 5e^t = 5e^{5e^t+t-5}$

Oct b. Problems (11-20)

11. $\frac{d}{dt}e^{5e^t-5-t^2} = e^{5e^t-5-t^2}\cdot(5e^t-2t)$ ~~rate~~

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12. $\frac{d}{dt}\left(\sin t+1\right) = \cos t$

13. $\frac{d}{dt}\frac{5}{t^2} = 0 + 5\times -2t^{-3} \Rightarrow \frac{-10}{t^3}$

14. $\frac{d}{dt}\frac{5}{(t+1)^3} = 0 + 5\times\frac{d}{dx}\frac{1}{(t+1)^3} = \frac{-15}{(t+1)^4}$

$\frac{d}{dx}\frac{1}{(t+1)^3} = \frac{d}{dx}(t+1)^{-3} \Rightarrow -3(t+1)^{-4} = \frac{-3}{(t+1)^4}$

15. $\frac{d}{dx}|x-2| = \begin{cases}\frac{d}{dx}[x-2] & x>2\\ \frac{d}{dx}(-(x-2)) & x<2\end{cases} = \begin{cases}1 & x>2\\ -1 & x<2\end{cases}$

16. $\frac{d}{dx}|x|^3 = \begin{cases}\frac{d}{dx}(x^3) & x>0\\ \frac{d}{dx}(-x^3) & x<0\end{cases} = \begin{cases}3x^2 & x>0\\ -3x^2 & x<0\end{cases}$

17. $\frac{d}{dx}e^{|x+2|} = \begin{cases}\frac{d}{dx}e^{x+2} & x>-2\\ \frac{d}{dx}e^{-(x+2)} & x<-2\end{cases} = \begin{cases}e^{x+2} & x>-2\\ -e^{-x-2} & x<-2\end{cases}$

18. $\frac{d}{dt}e^{e^{3t-1}} = e^{e^{3t-1}}\cdot\frac{d}{dx}e^{3t-1}$

$= e^{e^{3t-1}}\cdot 3e^{3t}$

19. $\frac{d^2}{dt^2}e^{e^{3t-1}} = \frac{d}{dt}3e^{3t}\cdot e^{e^{3t}-1}$

$= 9e^{3t}\cdot 3e^{3t}\cdot e^{e^{3t}-1}$

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20. $\frac{d}{dt}\frac{1}{(1-3t)^4} = \frac{d}{dt}(1-3t)^{-4} = -4(1-3t)^{-5} \cdot -3$

$= \frac{+12}{(1-3t)^5}$

Q. 1c Problems (31-40)

21. $\frac{d}{dt}\left(\frac{1}{\left(1-\frac{t}{2}\right)^2}\right) = \frac{d}{dt}\left(1-\frac{t}{2}\right)^{-2}$

$= -2\left(1-\frac{t}{2}\right)^{-3} \times -\frac{1}{2}$

$= \frac{1}{\left(1-\frac{t}{2}\right)^3}$

22. $\frac{d}{dx}e^{2x^2-5x+3} = \left(e^{2x^2-5x+3}\right)(4x-5)$

23. $\frac{d^2}{dx^2}e^{2x^2-5x+3} = \frac{d}{dx}(4x-5)\left(e^{2x^2-5x+3}\right)$

$4e^{2x^2-5x+3} + (4x-5)^2 e^{2x^2-5x+3}$

24. $\frac{d}{dx}(2x^5+x^3+8x)^4 = 4(2x^5+x^3+8x)^3 (10x^4+3x^2+8)$

25. $\frac{d}{dy}\frac{y^2}{6}e^{-3y} = \frac{y^2}{6}\frac{d}{dy}e^{-3y} + e^{-3y}\frac{d}{dy}\frac{y^2}{6}$

$= \frac{2y}{\cancel{6}_3}e^{-3y} + -\frac{y^2}{\cancel{6}_2}e^{-3y}(\cancel{3})$

$= \frac{y}{3}e^{-3y} - \frac{y^2}{2}e^{-3y}$

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26. $\frac{d^2}{dy^2}\frac{y^2}{6}e^{-3y} = \frac{d}{dy}\left(\frac{y}{3}e^{-3y} - \frac{y^2}{2}e^{-3y}\right)$

$= \frac{d}{dy}\left(\frac{y}{3}e^{-3y}\right) - \frac{d}{dy}\left(\frac{y^2}{2}e^{-3y}\right)$

$= \frac{d}{dy}e^{-3y}\times\left(\frac{y}{3} - \frac{y^2}{2}\right)$

$= \cancel{\frac{d}{dy}}\, e^{-3y}\left(\frac{1}{3} - 1\right) + -3\left(\frac{y}{3} - \frac{y^2}{2}\right)e^{-3y}$

$= -\frac{2}{3}e^{-3y} - 3e^{-3y}\left(\frac{y}{3} - \frac{y^2}{2}\right)$

27. $\frac{d}{dx}(3x^2+4)^5 = 5(3x^2+4)^4 \cdot 6x$

$= 30x\,(3x^2+4)^4$

28. $\frac{d}{dt}(2t^2+1)\cdot e^{5t} = e^{5t}\cdot 4t + (2t^2+1)\cdot 5\cdot e^{5t}$

$= 4t\,e^{5t} + 5e^{5t}(2t^2+1)$

29. $\frac{d}{dx}\left(\frac{e^{-x}}{x}\right) = \frac{1}{x}(-1)e^{-x} + e^{-x}(-1)\times\frac{1}{x^2}$

$= -\frac{e^{-x}}{x} - \frac{e^{-x}}{x^2}$

30. $\frac{d}{dz}(\ln z) = \frac{1}{z}$

0.1d Problems (31-40)

31. $\frac{d}{dy}\ln(2y+5) = \frac{2}{2y+5}$

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32. $\frac{d}{dt} \ln(t^2) = \frac{2t}{t^2} = \frac{2}{t}$

33. $\frac{d}{dx} e^{-x^2/2} = -x\, e^{-x^2/2}$

34. $\frac{d}{dx} \frac{x^3+5}{(2+x^3)^4} = \frac{1}{(2+x^3)^4} \times 3x^2 + \frac{(x^3+5).(-4)(3x^2)}{(2+x^3)^5}$

$= \frac{3x^2}{(2+x^3)^4} - \frac{12x^2(x^3+5)}{(2+x^3)^5}$

35. $\frac{d}{dx} \frac{2x+e^x+3}{e^{2x}-x} = (2x+e^x+3) \times \frac{(2e^{2x}-1).(-1)}{(e^{2x}-x)^2}$

$+ \frac{2+e^x}{e^{2x}-x}$

36. $\frac{d}{ds} s^3.(3s^2-4s+5)^4 = 3s^2(3s^2-4s+5)^4 +$
$s^3.4(3s^2-4s+5)(6s-4)$

37: $\frac{d}{dt} (2t^3-1)^2.(3t+7)^4$

$=(2t^3-1)^2.4(3t+7)^3.(3) + (3t+7)^4\, 2(2t^3-1)$
$(6t^2)$

38. $\frac{d}{dx} \frac{(x^3+1)^4}{(3x^2-5)^7} = \frac{(x^3+1)^4.(-7)(6x)}{(3x^2-5)^8} + \frac{4(x^3+1)^3.3x^2}{(3x^2-5)^7}$

39. $\frac{d}{dz}\left[(2z-(z^5+z^3)^6\right]^3 = 3(2z-(z^5+z^3)^6)^2$
$(2-6(z^5+z^3)^5(5z^4+3z^2))$

40. $\frac{d}{du} 2u.e^{-2u^2-5u} = 2e^{-2u^2-5u} + 2u.e^{-2u^2-5u}.$
$(-4u-5)$

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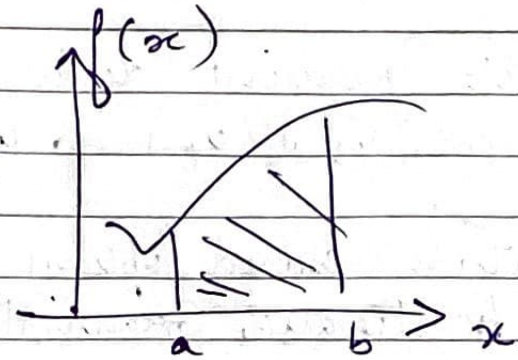

$$\int_a^b f(x)\,dx = \text{area under the curve}$$

$$\frac{d}{dx}F(x) = f(x)$$

- Derivatives & integrals are inverse operaⁿˢ

Eg. $\int_0^5 x\,dx = \frac{x^2}{2} = \frac{5^2}{2} = \frac{25}{2} = 12.5$

$$\boxed{\int_a^b x^n\,dx = \left.\frac{x^{n+1}}{n+1}\right|_a^b}$$

- Common formulas -

→ $\int a\,dx = ax + c$

→ $\int x\,dx = \frac{x^2}{2} + c$

→ $\int x^n\,dx = \frac{x^{n+1}}{n+1} + c \quad , n \neq -1$

→ $\int e^{bx}\,dx = \frac{1}{b}e^{bx} + c$

→ $\int a^x\,dx = \int e^{x\ln a}\,dx = \frac{1}{\ln a}a^x + c$

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- Use chain rule –

$\rightarrow \frac{d}{dx} e^{x^2} = e^{x^2} \cdot 2x$

$\rightarrow \frac{d}{dx} (x^2+3)^5 = 5(x^2+3)^4 \cdot 2x$

- Some other formulas –

$\rightarrow \int \frac{dx}{x} = \ln x + c$

$\rightarrow \int c f(x) dx = c \int f(x) dx$

$\rightarrow \int [f(x) + g(x)] dx = \int f(x) dx + \int g(x) dx$

$\rightarrow \int \cos x \, dx = \sin x + c$

$\rightarrow \int \sin x \, dx = -\cos x + C$

0.3 Integration by Parts

- Integra^n by parts is doing the product rule backwards.

$$\frac{d}{dx} uv = u \cdot \frac{dv}{dx} + v \cdot \frac{du}{dx}$$

$$\int d(uv) = uv = \int u \, dv + \int v \, du$$

$$\int u \cdot dv = uv - \int v \, du$$

Eg. $\int x e^x dx = x e^x - \int e^x dx$
$= x e^x - e^x + C$

Eg. $\int x^2 e^{2x} \Rightarrow x^2 \cdot \frac{1}{2} e^{2x} - \int \frac{1}{2} e^{2x} \, 2x \, dx$ --

- The gamma trick → If we have a definite integral from 0 to ∞, we often can skip using integra" by parts.

$b>0$, a is an integer, then

$$\boxed{\int_0^\infty x^a \cdot e^{-bx}\, dx = \frac{a!}{b^{a+1}}}$$

$$\int_0^\infty x^2 \cdot e^{-2x}\, dx = \frac{2!}{2^3} = \frac{2}{2^3} = \frac{1}{2^2} = \frac{1}{4}$$

- It is also related to $E[X^a]$ when X is an exponential RV as well as the density of a gamma R.V.